Superconservative Hilbert bases are unconditional for constant coefficients

Candidate partial result for arXiv:2001.01226

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Status. Candidate substantial partial result, likely valid; human review is needed. The unrestricted Markushevich-basis question remains open.

1 Source question and definitions

Berasategui, Berná, and Lassalle ask in Remark 2.5 of *Strong partially greedy bases and Lebesgue-type inequalities* [1] whether every superconservative basis is unconditional for constant coefficients.

For a finite set $A \subset \mathbb{N}$ and unimodular scalars $\varepsilon = (\varepsilon_n)_{n\in A}$, write

$$\mathbf{1}_{\varepsilon A} := \sum_{n\in A} \varepsilon_n x_n.$$

A semi-normalized Markushevich basis is Δ-superconservative when

$$\|\mathbf{1}_{\varepsilon A}\| \le \Delta \, \|\mathbf{1}_{\delta B}\| \tag{1.1}$$

whenever $A < B$ (that is, $\max A < \min B$), $|A| \le |B|$, and ε, δ are arbitrary signs. It is unconditional for constant coefficients when

$$\sup_{A\subset B} \frac{\|\mathbf{1}_{\varepsilon A}\|}{\|\mathbf{1}_{\varepsilon B}\|} < \infty. \tag{1.2}$$

Superdemocracy is stronger: the same comparison holds for arbitrary $|A| \le |B|$ and independent signs on the two sets.

2 Main partial result

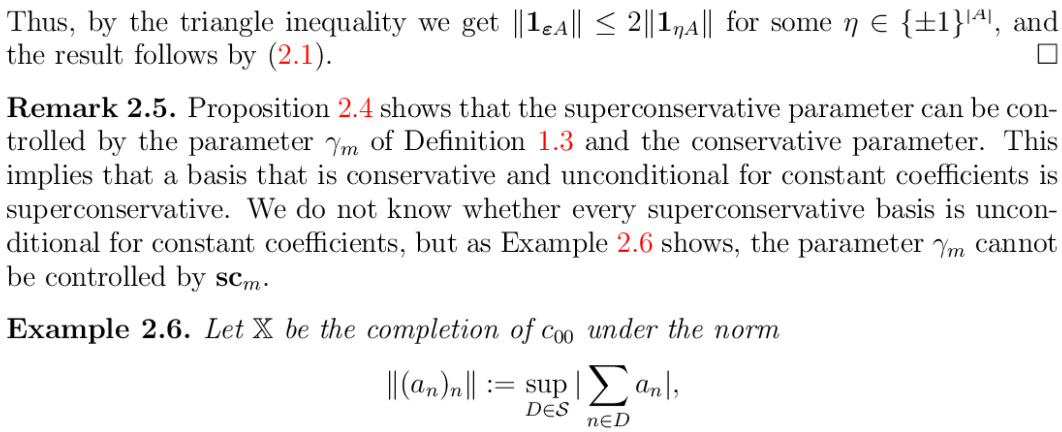
Theorem 1 (Hilbert-space affirmative answer)**.** *Let (x_n) be a semi-normalized Schauder basis of a Hilbert space H. Set*

$$a := \inf_n \|x_n\| > 0, \qquad b := \sup_n \|x_n\| < \infty, \qquad K := \sup_N \|P_N\|,$$

where P_N is the Nth partial-sum projection. If (x_n) is Δ-superconservative, then it is superdemocratic. Quantitatively, for all finite $A, B \subset \mathbb{N}$ with $|A| \le |B|$ and all signs,

$$\|\mathbf{1}_{\varepsilon A}\| \le \sqrt{3}\, \Delta^2 (1+K) \frac{b}{a} \, \|\mathbf{1}_{\delta B}\| . \tag{2.1}$$

In particular, the basis is unconditional for constant coefficients, with the same upper bound for its constant.



Thus, by the triangle inequality we get $\|\mathbf{1}_{\varepsilon A}\| \leq 2\|\mathbf{1}_{\eta A}\|$ for some $\eta \in \{\pm 1\}^{|A|}$, and the result follows by (2.1). ☐

Remark 2.5. Proposition 2.4 shows that the superconservative parameter can be controlled by the parameter γ_m of Definition 1.3 and the conservative parameter. This implies that a basis that is conservative and unconditional for constant coefficients is superconservative. We do not know whether every superconservative basis is unconditional for constant coefficients, but as Example 2.6 shows, the parameter γ_m cannot be controlled by $\mathbf{sc}_m$.

Example 2.6. *Let $\mathbb{X}$ be the completion of c_{00} under the norm*

$$\|(a_n)_n\| := \sup_{D\in\mathcal{S}} |\sum_{n\in D} a_n|,$$

Figure 1: Remark 2.5 in the source paper [1], PDF page 7.

Remark 2 (Weaker tail hypothesis)**.** *The proof does not require boundedness of P_N on arbitrary vectors. It is enough to assume bounded tail suppression on signed constant-coefficient vectors:*

$$K_{cc}^{\text{tail}} := \sup_{N,E,\delta} \frac{\left\|\mathbf{1}_{\delta E\cap(N,\infty)}\right\|}{\|\mathbf{1}_{\delta E}\|} < \infty. \tag{2.2}$$

Then (2.1) holds with $1+K$ replaced by K_{cc}^{tail}.

3 Idea of the proof

Hilbert-space randomization produces both sides of the same square-root scale. On a set of r basis vectors, some signing has norm at most $b\sqrt{r}$, while some signing has norm at least $a\sqrt{r}$. Superconservativeness transports the small signing from a later set to give an upper bound for an arbitrary numerator.

For an arbitrary prescribed denominator, split its support into ordered first and last halves. A large random signing of the first half transfers, by superconservativeness, to a lower bound for the prescribed signing on the last half. The Schauder tail operator then prevents that last half from being hidden by cancellation with the first half. This is exactly the step that is unavailable for a general Markushevich basis; hence the original question remains open outside the stated subcase.

4 Proof

Proof of Theorem 1. Let A be finite, $|A| = r$, and fix signs ε on A. Choose a set $C > A$ with $|C| = r$. For independent Rademacher signs $(\eta_n)_{n\in C}$, the Hilbert inner product and vanishing of the cross terms give

$$\mathbb{E}\left\|\sum_{n\in C}\eta_n x_n\right\|^2 = \sum_{n\in C}\|x_n\|^2 \leq b^2 r. \tag{4.1}$$

Consequently some signing η satisfies $\|\mathbf{1}_{\eta C}\| \leq b\sqrt{r}$. Applying (1.1) to $A < C$ gives

$$\|\mathbf{1}_{\varepsilon A}\| \leq \Delta b\sqrt{r}. \tag{4.2}$$

Now fix a nonempty finite set $B = \{b_1 < \cdots < b_s\}$ and prescribed signs δ on B. For $s \geq 2$, set $q = \lfloor s/2 \rfloor$ and

$$D = \{b_1, \ldots, b_q\}, \qquad E = \{b_{s-q+1}, \ldots, b_s\}.$$

Then $D < E$ and $|D| = |E| = q$. Applying (4.1) with the reverse inequality $\sum_{n \in D} \|x_n\|^2 \geq a^2 q$ shows that some signing η on D obeys

$$\|\mathbf{1}_{\eta D}\| \geq a\sqrt{q}.$$

Superconservativeness, now with the prescribed restriction of δ on E, yields

$$\|\mathbf{1}_{\delta E}\| \geq \frac{a\sqrt{q}}{\Delta}. \tag{4.3}$$

Put $N = b_{s-q}$. Since E is exactly the portion of B after N,

$$\mathbf{1}_{\delta E} = (I - P_N)\mathbf{1}_{\delta B}.$$

Thus $\|I - P_N\| \leq 1 + K$ and (4.3) imply

$$\|\mathbf{1}_{\delta B}\| \geq \frac{a\sqrt{q}}{\Delta(1+K)}. \tag{4.4}$$

Since $q = \lfloor s/2 \rfloor \geq s/3$, equations (4.2)–(4.4) and $r \leq s$ prove (2.1). If $s = 1$, the estimate follows directly from $a \leq \|x_n\| \leq b$ and is covered by the displayed constant. The proof with (2.2) is identical. $\square$

5 Type/cotype extension

Corollary 3. *Let X have Rademacher type p with constant T_p and cotype q with constant C_q, where $1 \leq p \leq 2 \leq q < \infty$. Let (x_n) be a semi-normalized Δ-superconservative Schauder basis with parameters a, b, K as in Theorem 1. Then, for $m \geq 2$, its constant-coefficient suppression parameters satisfy*

$$\gamma_m \leq 3^{1/q}\Delta^2(1+K)T_p C_q \frac{b}{a}\, m^{1/p-1/q}. \tag{5.1}$$

The same bound holds with $1 + K$ replaced by (2.2).

Proof. Type p gives a signing on a later r-set of norm at most $T_p b r^{1/p}$, so (4.2) becomes

$$\|\mathbf{1}_{\varepsilon A}\| \leq \Delta T_p b r^{1/p}.$$

Cotype q gives a signing on the first $q_0 = \lfloor s/2 \rfloor$ indices of norm at least $a q_0^{1/q}/C_q$. Repeating (4.3)–(4.4) yields

$$\|\mathbf{1}_{\delta B}\| \geq \frac{a q_0^{1/q}}{\Delta(1+K)C_q}.$$

Use $q_0 \geq s/3$, $r \leq s \leq m$, and divide. $\square$

6 Verification, upgrade attempts, and limitations

The proof was checked against all quantifiers in Definition 1.6 of the source: (1.1) permits independent signs and unequal cardinalities, while the proof only uses separated sets of equal size. The only operator identity is $\mathbf{1}_{\delta E} = (I - P_N)\mathbf{1}_{\delta B}$, valid because E is the final ordered half of B. The factor $\sqrt{3}$ comes from $\lfloor s/2 \rfloor \geq s/3$ for every $s \geq 2$. No computational or unproved dependency is used.

Eight focused upgrades tested: a direct reversal of the source's Proposition 2.4; ordered padding; removal of the Schauder hypothesis; the weaker constant-coefficient tail condition; type/cotype replacement of Hilbert randomization; repeated splitting; orthogonal sums of increasingly ill-conditioned finite Hilbert blocks; and nested-prefix low-eigenvalue constructions. The block constructions violate superconservativeness when a later block again has a small signed sum. Nested-prefix constructions leave exactly the uncontrolled head–tail cancellation measured by (2.2). The detailed audit is in the associated attempt note.

The result does *not* settle the source question for arbitrary Markushevich bases. It also does not settle the separate question in Section 5 of the source asking whether every partially greedy Markushevich basis is quasi-greedy. In the source's constant-one proof, the corresponding induction for a constant $C > 1$ can multiply C at each step and gives no uniform quasi-greedy bound.

Cheap run indexes and bounded searches through 12 August 2026 for the arXiv id, the quoted superconservative/UCC question, and the partially-greedy Markushevich question found no later paper explicitly resolving either full problem. Later work on strong partially greedy bases with arbitrary sequences does not supply the missing unrestricted implication. Novelty confidence is therefore moderate: the proof here is self-contained, but the literature search was bounded rather than exhaustive.

7 Human-review recommendation

Review the ordered-halves use of (1.1) and the tail identity preceding (4.4). If these are accepted, retain Theorem 1 and Corollary 3 as a substantial partial result. Do not upgrade the packet to a full solution unless (2.2) is derived from superconservativeness alone or an independent argument controls the missing head–tail cancellation.

References

[1] M. Berasategui, P. M. Berná, and S. Lassalle, *Strong partially greedy bases and Lebesgue-type inequalities*, J. Math. Anal. Appl. 492 (2020), 124410; arXiv:2001.01226v2.